

Computer Lab Assignments

72. In this problem you are led through the commands in *Mathematica* that enable you to obtain the symbolic Laplace transform of a differential equation and the solution of the initial-value problem by finding the inverse transform. In *Mathematica* the Laplace transform of a function $y(t)$ is obtained using `LaplaceTransform [y[t], t, s]`. In line two of the syntax we replace `LaplaceTransform [y[t], t, s]` by the symbol **Y**. (If you do not have *Mathematica*, then adapt the given procedure by finding the corresponding syntax for the CAS you have on hand.)

Consider the initial-value problem

$$y'' + 6y' + 9y = t \sin t, \quad y(0) = 2, \quad y'(0) = -1.$$

Load the Laplace transform package. Precisely reproduce and then, in turn, execute each line in the following sequence of commands. Either copy the output by hand or print out the results.

```
diffequat = y''[t] + 6y'[t] + 9y[t] == t Sin[t]
transformdeq = LaplaceTransform [diffequat, t, s] /.
{y[0] -> 2, y'[0] -> -1,
 LaplaceTransform [y[t], t, s] -> Y}
soln = Solve[transformdeq, Y]//Flatten
Y = Y/soln
InverseLaplaceTransform[Y, s, t]
```

73. Appropriately modify the procedure of Problem 72 to find a solution of

$$y''' + 3y' - 4y = 0, \\ y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1.$$

74. The charge $q(t)$ on a capacitor in an *LC*-series circuit is given by

$$\frac{d^2 q}{dt^2} + q = 1 - 4\mathcal{U}(t - \pi) + 6\mathcal{U}(t - 3\pi), \\ q(0) = 0, \quad q'(0) = 0.$$

Appropriately modify the procedure of Problem 72 to find $q(t)$. Graph your solution.

7.5 The Dirac Delta Function

INTRODUCTION In the last paragraph on page 284, we indicated that as an immediate consequence of Theorem 7.1.3, $F(s) = 1$ cannot be the Laplace transform of a function f that is piecewise continuous on $[0, \infty)$ and of exponential order. In the discussion that follows we are going to introduce a function that is very different from the kinds that you have studied in previous courses. We shall see that there does indeed exist a function—or, more precisely, a *generalized function*—whose Laplace transform is $F(s) = 1$.

UNIT IMPULSE Mechanical systems are often acted on by an external force (or electromotive force in an electrical circuit) of large magnitude that acts only for a very short period of time. For example, a vibrating airplane wing could be struck by lightning, a mass on a spring could be given a sharp blow by a ball peen hammer, and a ball (baseball, golf ball, tennis ball) could be sent soaring when struck violently by some kind of club (baseball bat, golf club, tennis racket). See Figure 7.5.1. The graph of the piecewise-defined function

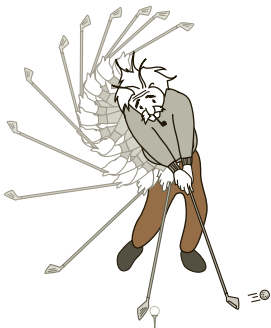


FIGURE 7.5.1 A golf club applies a force of large magnitude on the ball for a very short period of time

$$\delta_a(t - t_0) = \begin{cases} 0, & 0 \leq t < t_0 - a \\ \frac{1}{2a}, & t_0 - a \leq t < t_0 + a \\ 0, & t \geq t_0 + a, \end{cases} \quad (1)$$

$a > 0$, $t_0 > 0$, shown in Figure 7.5.2(a), could serve as a model for such a force. For a small value of a , $\delta_a(t - t_0)$ is essentially a constant function of large magnitude that is “on” for just a very short period of time, around t_0 . The behavior of $\delta_a(t - t_0)$ as $a \rightarrow 0$ is illustrated in Figure 7.5.2(b). The function $\delta_a(t - t_0)$ is called a **unit impulse**, because it possesses the integration property $\int_0^\infty \delta_a(t - t_0) dt = 1$.

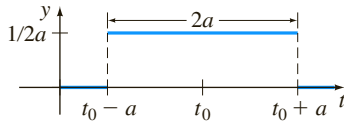
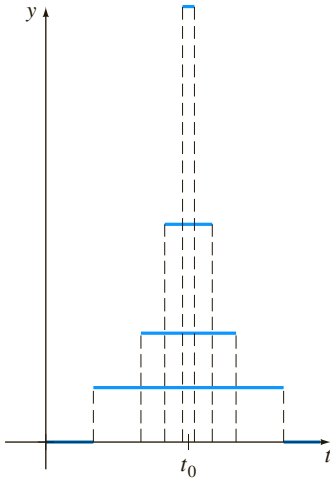
(a) graph of $\delta_a(t - t_0)$ (b) behavior of δ_a as $a \rightarrow 0$

FIGURE 7.5.2 Unit impulse

DIRAC DELTA FUNCTION In practice it is convenient to work with another type of unit impulse, a “function” that approximates $\delta_a(t - t_0)$ and is defined by the limit

$$\delta(t - t_0) = \lim_{a \rightarrow 0} \delta_a(t - t_0). \quad (2)$$

The latter expression, which is not a function at all, can be characterized by the two properties

$$(i) \delta(t - t_0) = \begin{cases} \infty, & t = t_0 \\ 0, & t \neq t_0 \end{cases} \quad \text{and} \quad (ii) \int_0^{\infty} \delta(t - t_0) dt = 1.$$

The unit impulse $\delta(t - t_0)$ is called the **Dirac delta function**.

It is possible to obtain the Laplace transform of the Dirac delta function by the formal assumption that $\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\}$.

THEOREM 7.5.1 Transform of the Dirac Delta Function

$$\text{For } t_0 > 0, \quad \mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}. \quad (3)$$

PROOF To begin, we can write $\delta_a(t - t_0)$ in terms of the unit step function by virtue of (11) and (12) of Section 7.3:

$$\delta_a(t - t_0) = \frac{1}{2a} [\mathcal{U}(t - (t_0 - a)) - \mathcal{U}(t - (t_0 + a))].$$

By linearity and (14) of Section 7.3 the Laplace transform of this last expression is

$$\mathcal{L}\{\delta_a(t - t_0)\} = \frac{1}{2a} \left[\frac{e^{-s(t_0 - a)}}{s} - \frac{e^{-s(t_0 + a)}}{s} \right] = e^{-st_0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right). \quad (4)$$

Since (4) has the indeterminate form $0/0$ as $a \rightarrow 0$, we apply L'Hôpital's Rule:

$$\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\} = e^{-st_0} \lim_{a \rightarrow 0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right) = e^{-st_0}. \quad \blacksquare$$

Now when $t_0 = 0$, it seems plausible to conclude from (3) that

$$\mathcal{L}\{\delta(t)\} = 1.$$

The last result emphasizes the fact that $\delta(t)$ is not the usual type of function that we have been considering, since we expect from Theorem 7.1.3 that $\mathcal{L}\{f(t)\} \rightarrow 0$ as $s \rightarrow \infty$.

EXAMPLE 1 Two Initial-Value Problems

Solve $y'' + y = 4\delta(t - 2\pi)$ subject to

$$(a) y(0) = 1, \quad y'(0) = 0 \quad (b) y(0) = 0, \quad y'(0) = 0.$$

The two initial-value problems could serve as models for describing the motion of a mass on a spring moving in a medium in which damping is negligible. At $t = 2\pi$ the mass is given a sharp blow. In (a) the mass is released from rest 1 unit below the equilibrium position. In (b) the mass is at rest in the equilibrium position.

SOLUTION (a) From (3) the Laplace transform of the differential equation is

$$s^2 Y(s) - s + Y(s) = 4e^{-2\pi s} \quad \text{or} \quad Y(s) = \frac{s}{s^2 + 1} + \frac{4e^{-2\pi s}}{s^2 + 1}.$$

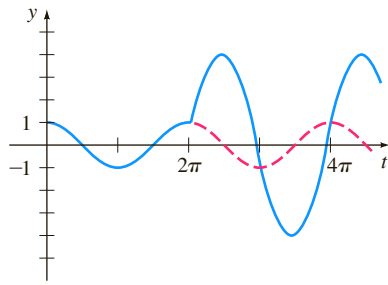


FIGURE 7.5.3 Mass is struck at $t = 2\pi$ in part (a) of Example 1

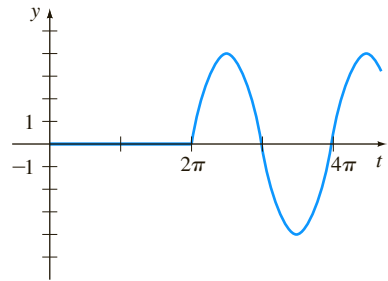


FIGURE 7.5.4 No motion until mass is struck at $t = 2\pi$ in part (b) of Example 1

Using the inverse form of the second translation theorem, (15) of Section 7.3, we find

$$y(t) = \cos t + 4 \sin(t - 2\pi) \mathcal{U}(t - 2\pi).$$

Since $\sin(t - 2\pi) = \sin t$, the foregoing solution can be written as

$$y(t) = \begin{cases} \cos t, & 0 \leq t < 2\pi \\ \cos t + 4 \sin t, & t \geq 2\pi. \end{cases} \quad (5)$$

In Figure 7.5.3 we see from the graph of (5) that the mass is exhibiting simple harmonic motion until it is struck at $t = 2\pi$. The influence of the unit impulse is to increase the amplitude of vibration to $\sqrt{17}$ for $t > 2\pi$.

(b) In this case the transform of the equation is simply

$$Y(s) = \frac{4e^{-2\pi s}}{s^2 + 1},$$

and so

$$\begin{aligned} y(t) &= 4 \sin(t - 2\pi) \mathcal{U}(t - 2\pi) \\ &= \begin{cases} 0, & 0 \leq t < 2\pi \\ 4 \sin t, & t \geq 2\pi. \end{cases} \end{aligned} \quad (6)$$

The graph of (6) in Figure 7.5.4 shows, as we would expect from the initial conditions that the mass exhibits no motion until it is struck at $t = 2\pi$. ■

REMARKS

(i) If $\delta(t - t_0)$ were a function in the usual sense, then property (i) on page 319 would imply $\int_0^\infty \delta(t - t_0) dt = 0$ rather than $\int_0^\infty \delta(t - t_0) dt = 1$. Because the Dirac delta function did not “behave” like an ordinary function, even though its users produced correct results, it was met initially with great scorn by mathematicians. However, in the 1940s Dirac’s controversial function was put on a rigorous footing by the French mathematician Laurent Schwartz in his book *Théorie des distributions*, and this, in turn, led to an entirely new branch of mathematics known as the **theory of distributions** or **generalized functions**. In this theory (2) is not an accepted definition of $\delta(t - t_0)$, nor does one speak of a function whose values are either ∞ or 0. Although we shall not pursue this topic any further, suffice it to say that the Dirac delta function is best characterized by its effect on other functions. If f is a continuous function, then

$$\int_0^\infty f(t) \delta(t - t_0) dt = f(t_0) \quad (7)$$

can be taken as the *definition* of $\delta(t - t_0)$. This result is known as the **sifting property**, since $\delta(t - t_0)$ has the effect of sifting the value $f(t_0)$ out of the set of values of f on $[0, \infty)$. Note that property (ii) (with $f(t) = 1$) and (3) (with $f(t) = e^{-st}$) are consistent with (7).

(ii) In (iii) in the *Remarks* at the end of Section 7.2 we indicated that the transfer function of a general linear n th-order differential equation with constant coefficients is $W(s) = 1/P(s)$, where $P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$. The transfer function is the Laplace transform of function $w(t)$, called the **weight function** of a linear system. But $w(t)$ can also be characterized in terms of the

discussion at hand. For simplicity let us consider a second-order linear system in which the input is a unit impulse at $t = 0$:

$$a_2 y'' + a_1 y' + a_0 y = \delta(t), \quad y(0) = 0, \quad y'(0) = 0.$$

Applying the Laplace transform and using $\mathcal{L}\{\delta(t)\} = 1$ shows that the transform of the response y in this case is the transfer function

$$Y(s) = \frac{1}{a_2 s^2 + a_1 s + a_0} = \frac{1}{P(s)} = W(s) \quad \text{and} \quad \text{so} \quad y = \mathcal{L}^{-1}\left\{\frac{1}{P(s)}\right\} = w(t).$$

From this we can see, in general, that the weight function $y = w(t)$ of an n th-order linear system is the zero-state response of the system to a unit impulse. For this reason $w(t)$ is also called the **impulse response** of the system.

EXERCISES 7.5

Answers to selected odd-numbered problems begin on page ANS-12.

In Problems 1–12 use the Laplace transform to solve the given initial-value problem.

1. $y' - 3y = \delta(t - 2), \quad y(0) = 0$
2. $y' + y = \delta(t - 1), \quad y(0) = 2$
3. $y'' + y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 1$
4. $y'' + 16y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 0$
5. $y'' + y = \delta(t - \frac{1}{2}\pi) + \delta(t - \frac{3}{2}\pi), \quad y(0) = 0, \quad y'(0) = 0$
6. $y'' + y = \delta(t - 2\pi) + \delta(t - 4\pi), \quad y(0) = 1, \quad y'(0) = 0$
7. $y'' + 2y' = \delta(t - 1), \quad y(0) = 0, \quad y'(0) = 1$
8. $y'' - 2y' = 1 + \delta(t - 2), \quad y(0) = 0, \quad y'(0) = 1$
9. $y'' + 4y' + 5y = \delta(t - 2\pi), \quad y(0) = 0, \quad y'(0) = 0$
10. $y'' + 2y' + y = \delta(t - 1), \quad y(0) = 0, \quad y'(0) = 0$
11. $y'' + 4y' + 13y = \delta(t - \pi) + \delta(t - 3\pi), \quad y(0) = 1, \quad y'(0) = 0$
12. $y'' - 7y' + 6y = e^t + \delta(t - 2) + \delta(t - 4), \quad y(0) = 0, \quad y'(0) = 0$

In Problems 13 and 14 use the Laplace transform to solve the given initial-value problem. Graph your solution on the interval $[0, 8\pi]$.

13. $y'' + y = \sum_{k=1}^{\infty} \delta(t - k\pi), \quad y(0) = 0, \quad y'(0) = 1$
14. $y'' + y = \sum_{k=1}^{\infty} \delta(t - 2k\pi), \quad y(0) = 0, \quad y'(0) = 1$

In Problems 15 and 16 a uniform beam of length L carries a concentrated load w_0 at $x = \frac{1}{2}L$. See Figure 7.5.5 (Problem 15) and Figure 7.5.6 (Problem 16). Use the Laplace transform to solve the differential equation

$$EI \frac{d^4 y}{dx^4} = w_0 \delta(x - \frac{1}{2}L), \quad 0 < x < L,$$

subject to the given boundary conditions.

15. $y(0) = 0, \quad y'(0) = 0, \quad y''(L) = 0, \quad y'''(L) = 0$

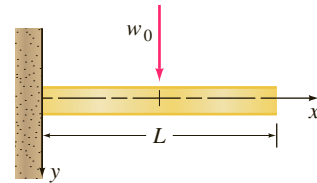


FIGURE 7.5.5 Beam embedded at its left end and free at its right end

16. $y(0) = 0, \quad y'(0) = 0, \quad y(L) = 0, \quad y'(L) = 0$

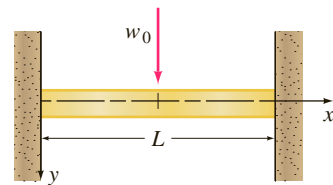


FIGURE 7.5.6 Beam embedded at both ends

Discussion Problems

17. Someone tells you that the solutions of the two IVPs

$$\begin{aligned} y'' + 2y' + 10y &= 0, & y(0) &= 0, & y'(0) &= 1 \\ y'' + 2y' + 10y &= \delta(t), & y(0) &= 0, & y'(0) &= 0 \end{aligned}$$

are exactly the same. Do you agree or disagree? Defend your answer.

18. Reread (i) in the *Remarks* at the end of this section. Then use the Laplace transform to solve the initial-value problem:

$$y'' + 4y' + 3y = e^t \delta(t - 1), \quad y(0) = 0, \quad y'(0) = 2.$$

Use a graphing utility to graph $y(t)$ for $0 \leq t \leq 5$.

7.6 Systems of Linear Differential Equations

INTRODUCTION When initial conditions are specified, the Laplace transform of each equation in a system of linear differential equations with constant coefficients reduces the system of DEs to a set of simultaneous algebraic equations in the transformed functions. We solve the system of algebraic equations for each of the transformed functions and then find the inverse Laplace transforms in the usual manner.

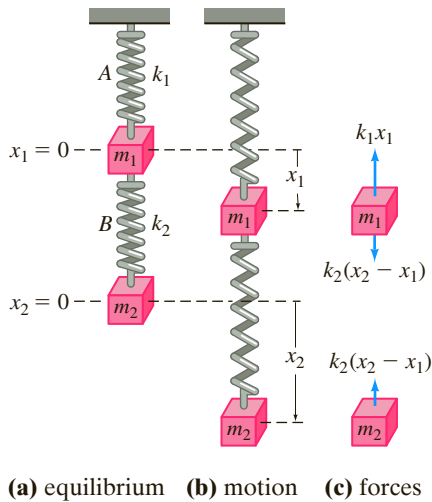


FIGURE 7.6.1 Coupled spring/mass system

COUPLED SPRINGS Two masses m_1 and m_2 are connected to two springs A and B of negligible mass having spring constants k_1 and k_2 , respectively. In turn the two springs are attached as shown in Figure 7.6.1. Let $x_1(t)$ and $x_2(t)$ denote the vertical displacements of the masses from their equilibrium positions. When the system is in motion, spring B is subject to both an elongation and a compression; hence its net elongation is $x_2 - x_1$. Therefore it follows from Hooke's law that springs A and B exert forces $-k_1x_1$ and $k_2(x_2 - x_1)$, respectively, on m_1 . If no external force is impressed on the system and if no damping force is present, then the net force on m_1 is $-k_1x_1 + k_2(x_2 - x_1)$. By Newton's second law we can write

$$m_1 \frac{d^2x_1}{dt^2} = -k_1x_1 + k_2(x_2 - x_1).$$

Similarly, the net force exerted on mass m_2 is due solely to the net elongation of B ; that is, $-k_2(x_2 - x_1)$. Hence we have

$$m_2 \frac{d^2x_2}{dt^2} = -k_2(x_2 - x_1).$$

In other words, the motion of the coupled system is represented by the system of simultaneous second-order differential equations

$$\begin{aligned} m_1 x_1'' &= -k_1x_1 + k_2(x_2 - x_1) \\ m_2 x_2'' &= -k_2(x_2 - x_1). \end{aligned} \quad (1)$$

In the next example we solve (1) under the assumptions that $k_1 = 6$, $k_2 = 4$, $m_1 = 1$, $m_2 = 1$, and that the masses start from their equilibrium positions with opposite unit velocities.

EXAMPLE 1 Coupled Springs

$$\begin{aligned} \text{Solve} \quad x_1'' + 10x_1 - 4x_2 &= 0 \\ -4x_1 + x_2'' + 4x_2 &= 0 \end{aligned} \quad (2)$$

subject to $x_1(0) = 0$, $x_1'(0) = 1$, $x_2(0) = 0$, $x_2'(0) = -1$.

SOLUTION The Laplace transform of each equation is

$$\begin{aligned} s^2X_1(s) - sx_1(0) - x_1'(0) + 10X_1(s) - 4X_2(s) &= 0 \\ -4X_1(s) + s^2X_2(s) - sx_2(0) - x_2'(0) + 4X_2(s) &= 0, \end{aligned}$$

where $X_1(s) = \mathcal{L}\{x_1(t)\}$ and $X_2(s) = \mathcal{L}\{x_2(t)\}$. The preceding system is the same as

$$\begin{aligned} (s^2 + 10)X_1(s) - 4X_2(s) &= 1 \\ -4X_1(s) + (s^2 + 4)X_2(s) &= -1. \end{aligned} \quad (3)$$

Solving (3) for $X_1(s)$ and using partial fractions on the result yields

$$X_1(s) = \frac{s^2}{(s^2 + 2)(s^2 + 12)} = -\frac{1/5}{s^2 + 2} + \frac{6/5}{s^2 + 12},$$

and therefore

$$\begin{aligned} x_1(t) &= -\frac{1}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} + \frac{6}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t. \end{aligned}$$

Substituting the expression for $X_1(s)$ into the first equation of (3) gives

$$X_2(s) = -\frac{s^2 + 6}{(s^2 + 2)(s^2 + 12)} = -\frac{2/5}{s^2 + 2} - \frac{3/5}{s^2 + 12}$$

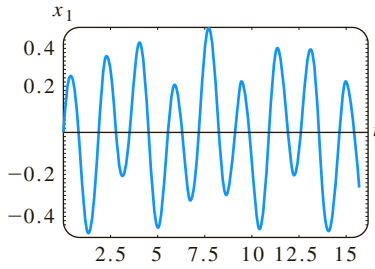
and

$$\begin{aligned} x_2(t) &= -\frac{2}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} - \frac{3}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned}$$

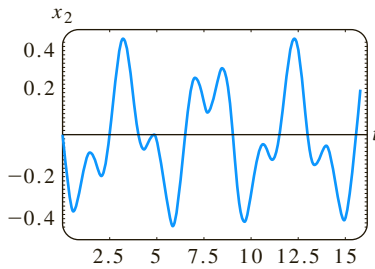
Finally, the solution to the given system (2) is

$$\begin{aligned} x_1(t) &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t \\ x_2(t) &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned} \quad (4)$$

The graphs of x_1 and x_2 in Figure 7.6.2 reveal the complicated oscillatory motion of each mass.



(a) plot of $x_1(t)$



(b) plot of $x_2(t)$

FIGURE 7.6.2 Displacements of the two masses in Example 1

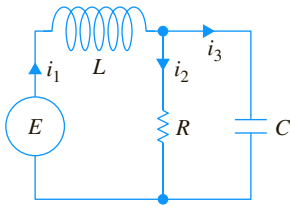


FIGURE 7.6.3 Electrical network

NETWORKS In (18) of Section 3.3 we saw the currents $i_1(t)$ and $i_2(t)$ in the network shown in Figure 7.6.3, containing an inductor, a resistor, and a capacitor, were governed by the system of first-order differential equations

$$\begin{aligned} L \frac{di_1}{dt} + Ri_2 &= E(t) \\ RC \frac{di_2}{dt} + i_2 - i_1 &= 0. \end{aligned} \quad (5)$$

We solve this system by the Laplace transform in the next example.

EXAMPLE 2 An Electrical Network

Solve the system in (5) under the conditions $E(t) = 60 \text{ V}$, $L = 1 \text{ h}$, $R = 50 \Omega$, $C = 10^{-4} \text{ f}$, and the currents i_1 and i_2 are initially zero.

SOLUTION We must solve

$$\begin{aligned} \frac{di_1}{dt} + 50i_2 &= 60 \\ 50(10^{-4}) \frac{di_2}{dt} + i_2 - i_1 &= 0 \end{aligned}$$

subject to $i_1(0) = 0$, $i_2(0) = 0$.

Applying the Laplace transform to each equation of the system and simplifying gives

$$\begin{aligned} sI_1(s) + 50I_2(s) &= \frac{60}{s} \\ -200I_1(s) + (s + 200)I_2(s) &= 0, \end{aligned}$$

where $I_1(s) = \mathcal{L}\{i_1(t)\}$ and $I_2(s) = \mathcal{L}\{i_2(t)\}$. Solving the system for I_1 and I_2 and decomposing the results into partial fractions gives

$$\begin{aligned} I_1(s) &= \frac{60s + 12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{60}{(s + 100)^2} \\ I_2(s) &= \frac{12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{120}{(s + 100)^2}. \end{aligned}$$

Taking the inverse Laplace transform, we find the currents to be

$$\begin{aligned} i_1(t) &= \frac{6}{5} - \frac{6}{5}e^{-100t} - 60te^{-100t} \\ i_2(t) &= \frac{6}{5} - \frac{6}{5}e^{-100t} - 120te^{-100t}. \end{aligned}$$

Note that both $i_1(t)$ and $i_2(t)$ in Example 2 tend toward the value $E/R = \frac{6}{5}$ as $t \rightarrow \infty$. Furthermore, since the current through the capacitor is $i_3(t) = i_1(t) - i_2(t) = 60te^{-100t}$, we observe that $i_3(t) \rightarrow 0$ as $t \rightarrow \infty$.

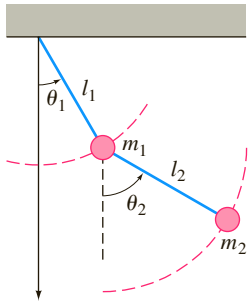


FIGURE 7.6.4 Double pendulum

DOUBLE PENDULUM Consider the double-pendulum system consisting of a pendulum attached to a pendulum shown in Figure 7.6.4. We assume that the system oscillates in a vertical plane under the influence of gravity, that the mass of each rod is negligible, and that no damping forces act on the system. Figure 7.6.4 also shows that the displacement angle θ_1 is measured (in radians) from a vertical line extending downward from the pivot of the system and that θ_2 is measured from a vertical line extending downward from the center of mass m_1 . The positive direction is to the right; the negative direction is to the left. As we might expect from the analysis leading to equation (6) of Section 5.3, the system of differential equations describing the motion is nonlinear:

$$\begin{aligned} (m_1 + m_2)l_1^2\theta_1'' + m_2l_1l_2\theta_2'' \cos(\theta_1 - \theta_2) + m_2l_1l_2(\theta_2')^2 \sin(\theta_1 - \theta_2) + (m_1 + m_2)l_1g \sin \theta_1 &= 0 \\ m_2l_2^2\theta_2'' + m_2l_1l_2\theta_1'' \cos(\theta_1 - \theta_2) - m_2l_1l_2(\theta_1')^2 \sin(\theta_1 - \theta_2) + m_2l_2g \sin \theta_2 &= 0. \end{aligned} \quad (6)$$

But if the displacements $\theta_1(t)$ and $\theta_2(t)$ are assumed to be small, then the approximations $\cos(\theta_1 - \theta_2) \approx 1$, $\sin(\theta_1 - \theta_2) \approx 0$, $\sin \theta_1 \approx \theta_1$, $\sin \theta_2 \approx \theta_2$ enable us to replace system (6) by the linearization

$$\begin{aligned} (m_1 + m_2)l_1^2\theta_1'' + m_2l_1l_2\theta_2'' + (m_1 + m_2)l_1g\theta_1 &= 0 \\ m_2l_2^2\theta_2'' + m_2l_1l_2\theta_1'' + m_2l_2g\theta_2 &= 0. \end{aligned} \quad (7)$$

EXAMPLE 3 Double Pendulum

It is left as an exercise to fill in the details of using the Laplace transform to solve system (7) when $m_1 = 3$, $m_2 = 1$, $l_1 = l_2 = 16$, $\theta_1(0) = 1$, $\theta_2(0) = -1$, $\theta_1'(0) = 0$, and $\theta_2'(0) = 0$. You should find that

$$\begin{aligned} \theta_1(t) &= \frac{1}{4} \cos \frac{2}{\sqrt{3}}t + \frac{3}{4} \cos 2t \\ \theta_2(t) &= \frac{1}{2} \cos \frac{2}{\sqrt{3}}t - \frac{3}{2} \cos 2t. \end{aligned} \quad (8)$$

With the aid of a CAS the positions of the two masses at $t = 0$ and at subsequent times are shown in Figure 7.6.5. See Problem 21 in Exercises 7.6.

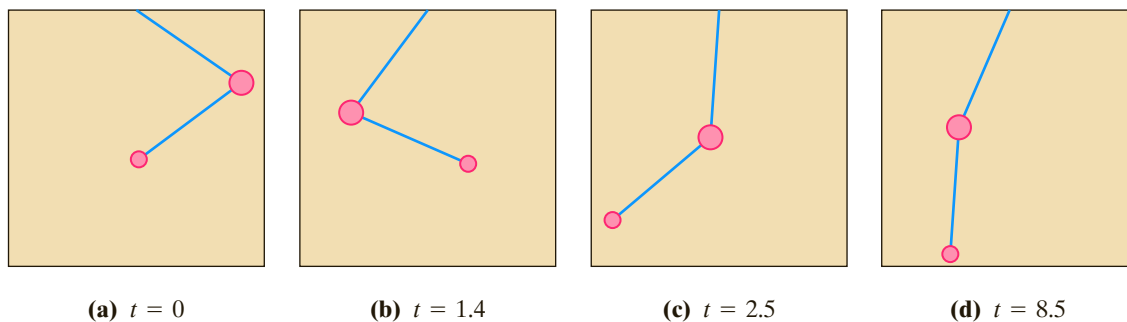


FIGURE 7.6.5 Positions of masses on double pendulum at various times in Example 3

EXERCISES 7.6

Answers to selected odd-numbered problems begin on page ANS-12.

In Problems 1–12 use the Laplace transform to solve the given system of differential equations.

1. $\frac{dx}{dt} = -x + y$
 $\frac{dy}{dt} = 2x$
 $x(0) = 0, \quad y(0) = 1$
2. $\frac{dx}{dt} = 2y + e^t$
 $\frac{dy}{dt} = 8x - t$
 $x(0) = 1, \quad y(0) = 1$
3. $\frac{dx}{dt} = x - 2y$
 $\frac{dy}{dt} = 5x - y$
 $x(0) = -1, \quad y(0) = 2$
4. $\frac{dx}{dt} + 3x + \frac{dy}{dt} = 1$
 $\frac{dx}{dt} - x + \frac{dy}{dt} - y = e^t$
 $x(0) = 0, \quad y(0) = 0$
5. $2\frac{dx}{dt} + \frac{dy}{dt} - 2x = 1$
 $\frac{dx}{dt} + \frac{dy}{dt} - 3x - 3y = 2$
 $x(0) = 0, \quad y(0) = 0$
6. $\frac{dx}{dt} + x - \frac{dy}{dt} + y = 0$
 $\frac{dx}{dt} + \frac{dy}{dt} + 2y = 0$
 $x(0) = 0, \quad y(0) = 1$
7. $\frac{d^2x}{dt^2} + x - y = 0$
 $\frac{d^2y}{dt^2} + y - x = 0$
 $x(0) = 0, \quad x'(0) = -2,$
 $y(0) = 0, \quad y'(0) = 1$
8. $\frac{d^2x}{dt^2} + \frac{dx}{dt} + \frac{dy}{dt} = 0$
 $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 4\frac{dx}{dt} = 0$
 $x(0) = 1, \quad x'(0) = 0,$
 $y(0) = -1, \quad y'(0) = 5$
9. $\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} = t^2$
 $\frac{d^2x}{dt^2} - \frac{d^2y}{dt^2} = 4t$
 $x(0) = 8, \quad x'(0) = 0,$
 $y(0) = 0, \quad y'(0) = 0$
10. $\frac{dx}{dt} - 4x + \frac{d^3y}{dt^3} = 6 \sin t$
 $\frac{dx}{dt} + 2x - 2\frac{d^3y}{dt^3} = 0$
 $x(0) = 0, \quad y(0) = 0,$
 $x'(0) = 0, \quad y''(0) = 0$
11. $\frac{d^2x}{dt^2} + 3\frac{dy}{dt} + 3y = 0$
 $\frac{d^2x}{dt^2} + 3y = te^{-t}$
 $x(0) = 0, \quad x'(0) = 2, \quad y(0) = 0$
12. $\frac{dx}{dt} = 4x - 2y + 2\mathcal{U}(t-1)$
 $\frac{dy}{dt} = 3x - y + \mathcal{U}(t-1)$
 $x(0) = 0, \quad y(0) = \frac{1}{2}$
13. Solve system (1) when $k_1 = 3, k_2 = 2, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x_1'(0) = 1, x_2(0) = 1, x_2'(0) = 0$.
14. Derive the system of differential equations describing the straight-line vertical motion of the coupled springs shown in Figure 7.6.6. Use the Laplace transform to solve the system when $k_1 = 1, k_2 = 1, k_3 = 1, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x_1'(0) = -1, x_2(0) = 0, x_2'(0) = 1$.

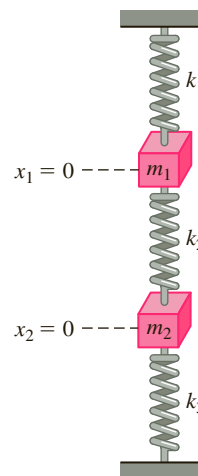


FIGURE 7.6.6 Coupled springs in Problem 14

15. (a) Show that the system of differential equations for the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in Figure 7.6.7 is

$$L_1 \frac{di_2}{dt} + Ri_2 + Ri_3 = E(t)$$

$$L_2 \frac{di_3}{dt} + Ri_2 + Ri_3 = E(t).$$

- (b) Solve the system in part (a) if $R = 5 \, \Omega, L_1 = 0.01 \, \text{h}, L_2 = 0.0125 \, \text{h}, E = 100 \, \text{V}, i_2(0) = 0,$ and $i_3(0) = 0$.
- (c) Determine the current $i_1(t)$.

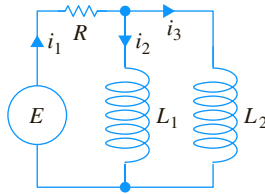


FIGURE 7.6.7 Network in Problem 15

16. (a) In Problem 12 in Exercises 3.3 you were asked to show that the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in Figure 7.6.8 satisfy

$$\begin{aligned} L \frac{di_2}{dt} + L \frac{di_3}{dt} + R_1 i_2 &= E(t) \\ -R_1 \frac{di_2}{dt} + R_2 \frac{di_3}{dt} + \frac{1}{C} i_3 &= 0. \end{aligned}$$

Solve the system if $R_1 = 10 \, \Omega$, $R_2 = 5 \, \Omega$, $L = 1 \, \text{h}$, $C = 0.2 \, \text{f}$,

$$E(t) = \begin{cases} 120, & 0 \leq t < 2 \\ 0, & t \geq 2, \end{cases}$$

$i_2(0) = 0$, and $i_3(0) = 0$.

- (b) Determine the current $i_1(t)$.

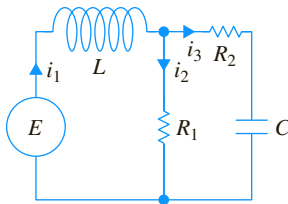


FIGURE 7.6.8 Network in Problem 16

17. Solve the system given in (17) of Section 3.3 when $R_1 = 6 \, \Omega$, $R_2 = 5 \, \Omega$, $L_1 = 1 \, \text{h}$, $L_2 = 1 \, \text{h}$, $E(t) = 50 \sin t \, \text{V}$, $i_2(0) = 0$, and $i_3(0) = 0$.
18. Solve (5) when $E = 60 \, \text{V}$, $L = \frac{1}{2} \, \text{h}$, $R = 50 \, \Omega$, $C = 10^{-4} \, \text{f}$, $i_1(0) = 0$, and $i_2(0) = 0$.
19. Solve (5) when $E = 60 \, \text{V}$, $L = 2 \, \text{h}$, $R = 50 \, \Omega$, $C = 10^{-4} \, \text{f}$, $i_1(0) = 0$, and $i_2(0) = 0$.
20. (a) Show that the system of differential equations for the charge on the capacitor $q(t)$ and the current $i_3(t)$ in the electrical network shown in Figure 7.6.9 is

$$\begin{aligned} R_1 \frac{dq}{dt} + \frac{1}{C} q + R_1 i_3 &= E(t) \\ L \frac{di_3}{dt} + R_2 i_3 - \frac{1}{C} q &= 0. \end{aligned}$$

- (b) Find the charge on the capacitor when $L = 1 \, \text{h}$, $R_1 = 1 \, \Omega$, $R_2 = 1 \, \Omega$, $C = 1 \, \text{f}$,

$$E(t) = \begin{cases} 0, & 0 < t < 1 \\ 50e^{-t}, & t \geq 1, \end{cases}$$

$i_3(0) = 0$, and $q(0) = 0$.

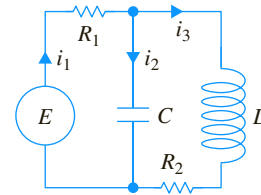


FIGURE 7.6.9 Network in Problem 20

Computer Lab Assignments

21. (a) Use the Laplace transform and the information given in Example 3 to obtain the solution (8) of the system given in (7).
- (b) Use a graphing utility to graph $\theta_1(t)$ and $\theta_2(t)$ in the $t\theta$ -plane. Which mass has extreme displacements of greater magnitude? Use the graphs to estimate the first time that each mass passes through its equilibrium position. Discuss whether the motion of the pendulums is periodic.
- (c) Graph $\theta_1(t)$ and $\theta_2(t)$ in the $\theta_1\theta_2$ -plane as parametric equations. The curve defined by these parametric equations is called a **Lissajous curve**.
- (d) The positions of the masses at $t = 0$ are given in Figure 7.6.5(a). Note that we have used $1 \, \text{radian} \approx 57.3^\circ$. Use a calculator or a table application in a CAS to construct a table of values of the angles θ_1 and θ_2 for $t = 1, 2, \dots, 10 \, \text{s}$. Then plot the positions of the two masses at these times.
- (e) Use a CAS to find the first time that $\theta_1(t) = \theta_2(t)$ and compute the corresponding angular value. Plot the positions of the two masses at these times.
- (f) Utilize the CAS to draw appropriate lines to simulate the pendulum rods, as in Figure 7.6.5. Use the animation capability of your CAS to make a “movie” of the motion of the double pendulum from $t = 0$ to $t = 10$ using a time increment of 0.1. [Hint: Express the coordinates $(x_1(t), y_1(t))$ and $(x_2(t), y_2(t))$ of the masses m_1 and m_2 , respectively, in terms of $\theta_1(t)$ and $\theta_2(t)$.]